TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	NanoGrid-AI: AI-enabled nanogrid and microgrid energy management platform with predictive digital twins for grid-resilient buildings and campuses			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Energy & Resources		Energy & Resources	
	Medium Category	Small Category	Medium Category	Small Category
	Power IT	Microgrid technology	Power IT	Intelligent power grid platform Technology
Concept of Proposed Technology	NanoGrid-AI targets multi-energy nanogrids and microgrids at building and campus scale (PV, stationary battery storage, EV charging, heat pumps, and controllable loads) that must operate reliably under grid constraints. The proposed technology is a modular software platform that combines: 1) Predictive digital twin models of the energy assets and building thermal dynamics (physics-based models calibrated with data, and data-driven surrogate models). TNO contributes model calibration and Bayesian inference tooling used in its Digital Built Environment work and living			

	labs [1-3]. 2) AI-based forecasting and diagnostics (load, PV, flexibility, and anomaly detection) using uncertainty quantification. Methods build on TNO and Wilkins-authored work on optimisation and uncertainty-aware prediction for electrified systems [4-6]. 3) Real-time optimisation and control (Model Predictive Control and mixed-integer optimisation) to schedule flexible assets to minimise peak import and operational cost while respecting comfort, safety and asset lifetime constraints. 4) Interoperability layer for data ingestion and control via open standards (e.g., IEC 61850 and OpenADR where applicable) and secure data sharing. TNO contributes core methods and validation capability via its Digital Twin and predictive twinning expertise [2-3], its Stieltjesweg living lab for climate control optimisation [1,7], and hardware-in-the-loop test facilities for energy systems integration (Heat Pump Application Center and Building Innovation Lab) [8-9].
Rationale for Joint R&D Project	Grid congestion and the rapid roll-out of distributed energy resources require advanced local energy management that can predict, optimise and verify system behaviour before deployment. Collaboration is required to combine TNO's predictive digital twin, optimisation and validation capabilities with the Korean SME partner's product platform, local market requirements, and deployment sites. The joint R&D will accelerate transfer from lab-validated algorithms to pilot-ready software integrated with commercial controllers and field data.
Key Words	1. Microgrid energy management 2. Predictive digital twin 3. Model predictive control 4. AI forecasting 5. Grid flexibility

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	Develop and demonstrate an AI-enabled nano/microgrid energy management system that: - Builds and maintains a predictive digital twin of the local energy system (electrical and thermal). - Provides probabilistic short-term forecasts (15 min to 48 h) for load and generation and identifies flexibility. - Optimises scheduling and real-time control of storage, EV charging and thermal systems under grid constraints. - Integrates secure data sharing, logging, and model lifecycle management for deployment at customer sites.			
R&D Target Specifications	Target performance (to be validated in pilot): - Peak grid import reduction: 15% to 25% versus baseline rule-based control. - Renewable self-consumption increase: 10% to 20% (PV + storage). - Forecast accuracy: Load and PV day-ahead MAE less than 5% of rated capacity; calibrated uncertainty intervals with 90% coverage. - Control cycle time: optimisation update every 5 minutes with fail-safe fallback. - Interoperability: integration with at least two controller/EMS interfaces (e.g., Modbus TCP, IEC 61850, or vendor APIs). - Cybersecurity: role-based access control, secure audit logging, and encrypted data-in-transit (TLS) on all external interfaces.			
R&D Plan	Planned duration: 36 months. **WP1 Requirements and data architecture (M1-M3):** use-case definition, data model, interface mapping, baseline KPIs. **WP2 Predictive digital twin development (M2-M10):** asset and thermal models, calibration workflow, model confidence assessment based on TNO digital twin practices [2-3]. **WP3 AI forecasting and diagnostics (M4-M12):** probabilistic forecasting, anomaly detection, uncertainty quantification inspired by TNO co-authored approaches to prediction and scheduling in electrified energy systems [4-6]. **WP4 Optimisation and control (M6-M18):** MPC and mixed-integer optimisation for storage, EV, and HVAC; degradation-aware constraints building on Wilkins and TNO work on charging optimisation under grid constraints [4-5]. **WP5 Platform integration and secure deployment (M8-M20):** containerised services, monitoring, model lifecycle management, security hardening.			

	WP6 Pilot demonstration and validation (M16-M28): deployment at a Korean SME provided site, A/B benchmarking vs baseline, iteration; test and validation methodology aligned with TNO's applied energy system test facilities [8-9]. **WP7 Exploitation and standardisation (M20-M36):** documentation, IP strategy, publication of non-confidential results, go-to-market plan.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Role of TNO (research institute):** • Define overall system architecture for the NanoGrid-AI platform and reference implementation. • Develop predictive digital twin models for electrical and thermal assets, including calibration and validation workflows. • Design AI-based forecasting and diagnostics algorithms with uncertainty quantification. • Develop optimisation and control strategies (e.g., MPC, mixed-integer optimisation) for flexible energy assets. • Integrate degradation-aware and constraint-aware control logic for storage, EV charging, and thermal systems. • Design data models and interfaces for secure ingestion, processing, and exchange of operational data. • Provide validation methodology and KPIs for performance, resilience, and energy efficiency. • Execute laboratory and hardware-in-the-loop validation using digital twin and energy system test facilities. • Support system verification through pilot testing and benchmarking against baseline operation. • Produce technical documentation, model descriptions, and validation reports to support deployment and future certification. • Contribute to scientific dissemination, standardisation activities, and methodological alignment with international best practices.
Expected R&R of Korean SME Partner	**Role of Korean SME partner:** • Define target application scenarios, system requirements, and deployment constraints for buildings or campuses. • Provide access to pilot sites, energy assets, and operational data streams (PV, storage, loads, EV chargers, etc.). • Supply or integrate local EMS, controllers, sensors, and communication interfaces. • Implement software interfaces required for data exchange with the NanoGrid-AI platform. • Support system integration, commissioning, and on-site validation activities. • Participate in performance evaluation, benchmarking, and iteration of control strategies. • Provide feedback on usability, scalability, and commercial feasibility of the developed solution. • Support compliance with local grid regulations, interconnection rules, and cybersecurity practices. • Lead commercialization planning, including productization, customer engagement, and

	market deployment strategy. • Drive post-project exploitation of results in commercial offerings and future product lines.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes: - A pilot-validated NanoGrid-AI software prototype (TRL 6) with documented APIs, deployment playbook, and monitoring dashboards. - Quantified performance evidence (peak reduction, cost, comfort, asset lifetime) through baseline comparison. - A reusable digital twin and optimisation library that can be ported to multiple microgrid and building contexts. - Joint IP (where appropriate) and scientific outputs. Impact: - Improved local energy autonomy and reduced grid congestion risk for buildings and campuses. - Faster commercial deployment of advanced EMS functionality by the Korean SME partner. - Strengthened Korea-EU collaboration on AI-enabled energy systems and digital twins.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization